Questions

Q1.

$$f(x) = (x^2 - 2)(2x - 3) - 21$$

(a) State the value of the remainder when f(x) is divided by $(2x - 3)$

(1)

(b) Use the factor theorem to show that $(x - 3)$ is a factor of f(x)

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(2)

(c) Hence,

(i) factorise f(x)

(ii) show that the equation f(x) = 0 has only one real root.

(5)

(Total for question = 8 marks)

(**Q04 WMA12/01, Oct 2021**)

Q2.

In this question you must show all stages of your working.

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Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of f(x),

(a) show that

$$a + 4b = -56$$

(2)

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Given also that when f(x) is divided by (x − 2) the remainder is −25

(b) find a second simplified equation linking a and b.

(2)

(c) Hence, using algebra and showing your working,

(i) find the value of a and the value of b,

(ii) fully factorise f(x).

(5)

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(Total for question = 9 marks)

(**Q04 WMA12/01, Oct 2023**)

Q3.

$$f(x) = 3x^3 + Ax^2 + Bx - 10$$

where A and B are integers.

Given that

- when f(x) is divided by (x – 1) the remainder is k
- when f(x) is divided by (x + 1) the remainder is –10k
- k is a constant

(a) show that

$$11A + 9B = 83$$

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(3)

Given also that (3x – 2) is a factor of f(x),

(b) find the value of A and the value of B.

(3)

(c) Hence find the quadratic expression g(x) such that

$$f(x) = (3x - 2)\, g(x)$$

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(2)

(Total for question = 8 marks)

(Q05 WMA12/01, Jan 2022)

Q4.

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$$f(x) = Ax^3 + 6x^2 - 4x + B$$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$
- $\int_3^5 f(x)dx = 176$

find the value of A and the value of B.

(Total for question = 7 marks)

(Q07 WMA12/01, June 2022)

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Q5.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

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(a) use the factor theorem to show that $a = -3$

(2)

(b) Hence show that the equation $f(x) = 0$ has only one real root.

(4)

(Total for question = 6 marks)

(Q02 WMA12/01, June 2023)

Q6.

$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51$$

(2)

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0$$

(2)

(c) Hence find the value of p and the value of q.

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(3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)\,g(x)$$

(2)

(Total for question = 9 marks)

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(**Q05 WMA12/01, Jan 2023**)

Q7.

$$f(x) = x^4 + x^3 + 2x^2 + ax + b,$$

where a and b are constants.

When f(x) is divided by (x – 1), the remainder is 7

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(a) Show that $a + b = 3$

(2)

When f(x) is divided by (x + 2), the remainder is –8

(b) Find the value of a and the value of b

(5)

(Total for question = 7 marks)

(**Q01 WMA12/01, Specimen papers**)

Q8.

$$f(x) = (x - 3)(3x^2 + x + a) - 35 \text{ where } a \text{ is a constant}$$

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(a) State the remainder when f(x) is divided by ($x - 3$).

(1)

Given ($3x - 2$) is a factor of f(x),

(b) show that $a = -17$

(2)

(c) Using algebra and showing each step of your working, fully factorise f(x).

(5)

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(Total for question = 8 marks)

(Q04 WMA12/01, Oct 2019)

Q9.

$$f(x) = ax^3 - x^2 + bx + 4$$

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where a and b are constants.

When $f(x)$ is divided by $(x + 4)$, the remainder is -108

(a) Use the remainder theorem to show that

$$16a + b = 24$$

(2)

Given also that $(2x - 1)$ is a factor of $f(x)$,

(b) find the value of a and the value of b.

(3)

(c) Find $f'(x)$.

(1)

(d) Hence find the exact coordinates of the stationary points of the curve with equation $y = f(x)$.

(4)

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(Total for question = 10 marks)

(Q03 WMA12/01, Oct 2020)

Q10.

$$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$$

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where a and b are constants.

When f(x) is divided by ($x + 1$), the remainder is 4

(a) Show that $a + b = -1$

(2)

When f(x) is divided by ($x - 2$) , the remainder is -23

(b) Find the value of a and the value of b.

(4)

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(Total for question = 6 marks)

(**Q01 WMA12/01, Jan 2021**)

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